Supplementary Material for

TraceCore: A Streaming Coreness Index over NetFlow for Time-Travel Forensics and Intrusion Detection

Wei Wu   Yuan Liu   Hui Lu   Fan Zhang   Zhihong Tian

This document supplements the main submission. To keep the main paper within the page limit, several result tables were moved here, together with a worked index example and the complete 40-run experiment ledger to which every result-table number in the paper traces. All values are reproduced verbatim from the same experiment runs as the main paper; no number changes between the paper and this supplement.

1 Worked Index Example (Three Epochs)

Suppose hosts $\{a, b, c\}$ have coreness $(1, 1, 2)$ at epoch t: $S_{t,1} = \{a, b\}$, $S_{t,2} = \{c\}$. At $t{+}1$, b rises to 2: `CoreDelta` appends $(t{+}1, b, 1, 2)$; `HostTimeline`$[b]$ appends $(t{+}1, 2)$; the epoch close materializes $S_{t+1,1} = \{a\}$ and $S_{t+1,2} = \{b, c\}$. At $t{+}2$ with no changes, $S_{t+2,\cdot}$ carries forward identically and `CoreDelta`/`HostTimeline` receive nothing — quiet epochs cost only the carry-forward.

2 Additional Result Tables

Table 1: Static-snapshot detection across four NF-UQ-NIDS-v2 sub-datasets (best per-dataset score). A single undirected graph per sub-dataset, hosts scored by raw k-shell / fanout composites, no temporal data; confirms the structural signal is not specific to the temporal corpus.

Sub-dataset	hosts	edges	pos.	best score	R@1%	R@5%
NF-BoT-IoT-v2	293	332	8	k_shell	1.000	1.000
NF-CSE-CIC-IDS2018-v2	255,042	1,866,047	648	k_shell	0.245	0.431
NF-ToN-IoT-v2	29,245	30,789	31	k_shell	0.774	0.774
NF-UNSW-NB15-v2	42	150	4	fanout	1.000	1.000

Table 2: Ablation on NF-CSE-CIC-IDS2018-v2 (18.8 M flows): the graph features carry the lift. TraceCore-only is essentially as good as TraceCore+Flow and well ahead of Flow-only on every PR-sensitive metric; subsampling training data 5× (frac 1.0→0.2) barely moves F1.

Config	F1	PR-AUC	R@10^{-4}	R@10^{-3}
TraceCore + Flow (frac=1.0)	0.9816	0.9983	0.9646	0.9680
TraceCore + Flow (frac=0.2)	0.9819	0.9986	0.9648	0.9674
TraceCore only (frac=0.2)	0.9795	0.9981	0.9602	0.9621
Flow only (frac=0.2)	0.9787	0.9791	0.9525	0.9609

Table 3: Streaming-build measurements on the 28-day LANL window (single CPU core). The query-load row runs a concurrent client issuing `point_coreness` queries against the partially-built index throughout the build. Epoch granularity, not core decomposition, is the cost driver.

Config	epochs	wall	flows/s	epoch p50/p99 (ms)	RSS / index
W=60 s	40,120	4,564 s	24.8 K	67.5 / 696.2	4.04 GB / 355 MB
W=300 s (bare)	8,024	1,884 s	60.2 K	(mean 235)	— / 142 MB
W=300 s + queries	8,024	4,245 s	26.7 K	546 / 1,378	2.14 GB / 142 MB
W=900 s	2,676	292 s	387.9 K	90.7 / 533.5	2.25 GB / 65 MB

Table 4: Top-10 LightGBM feature importances on NF-CSE-CIC-IDS2018-v2 (split-gain). The four TraceCore graph features (bold) account for 61.4% of the top-10 split-gain (sum 6,988 of 11,381) and occupy the top four slots.

Feature	Split gain	Origin
src_fanout_out	2,045	TraceCore graph
src_fanout_in	1,904	TraceCore graph
dst_fanout_in	1,563	TraceCore graph
dst_fanout_out	1,476	TraceCore graph
byte_ratio	989	flow
log_in_bytes	953	flow
dst_kshell	773	TraceCore graph
TCP_FLAGS	670	flow
log_out_bytes	608	flow
src_kshell	400	TraceCore graph

Table 5: Per-attack-class recall at FP= 1% on NF-ToN-IoT-v2: TraceCore+LightGBM (structural+flow features) versus Flow-only LightGBM. The structural-feature advantage is largest on attacks involving *co-ordinated multi-host* activity (MITM, ransomware, scanning); it is smallest on volumetric attacks already captured by per-flow rate features (DDoS, DoS) and absent on `Backdoor` where both classifiers already saturate above 0.99.

Attack class	#pos.	TraceCore	Flow-only	Δ
mitm	1,488	0.986	0.632	+35.4 pp
ransomware	659	0.991	0.672	+31.9 pp
scanning	756,607	0.957	0.825	+13.2 pp
injection	136,588	0.999	0.961	+3.8 pp
DoS	143,012	1.000	0.967	+3.3 pp
password	230,675	0.999	0.975	+2.5 pp
DDoS	404,881	0.999	0.976	+2.3 pp
xss	490,790	1.000	0.984	+1.6 pp
Backdoor	3,409	0.994	0.994	+0.0 pp

Table 6: Split-protocol variants at fraction 0.2 (single controlled run). The transductive variant (features from the full train+test graph) matches the standard inductive row to within 0.2 pp on every metric; the host-group split holds out 20% of source hosts entirely. This is the full table for the variants summarized in the main paper.

Dataset	Variant	PR-AUC	R@10^{-3}	ROC	F1@0.5
NF-CSE-CIC-IDS2018-v2	standard (inductive)	0.9987	0.9697	0.9998	0.9820
	transductive	0.9987	0.9698	0.9998	0.9820
	host-group	0.9228	0.9045	0.9619	0.2553
NF-ToN-IoT-v2	standard (inductive)	0.9985	0.9150	0.9970	0.9894
	transductive	0.9985	0.9171	0.9970	0.9894
	host-group	0.7103	0.0302	0.2318	0.0643
NF-UNSW-NB15-v2	standard (inductive)	0.9894	0.8805	0.9996	0.9599
	transductive	0.9892	0.8789	0.9996	0.9608
	host-group	0.9882	0.8041	0.9994	0.9579

Table 7: TraceCore-R residual sensitivity to the role count K (24-hour residual, R@FP $= 10^{-3}$ on the LANL red-team corpus). The strict-FP operating point is stable across an order of magnitude in K and collapses only at K=64. The EWMA half-life is similarly insensitive: R@10^{-3} stays in 0.738–0.768 across half-lives $\{6\text{h}, 1\text{d}, 4\text{d}\}$.

K	4	8	12	16	24	32	64
R@10^{-3}	0.768	0.768	0.787	0.768	0.756	0.756	0.579

3 Complete Experiment Ledger

Every result-table number in the main paper and in this supplement traces to one run `tag` in Table 8. The parsed per-run metrics and the derived-evidence file accompany this supplement; full per-run manifests and raw logs are part of the artifact release at publication.

Table 8: Complete experiment ledger: all 40 runs behind the results in the main paper and this supplement. Every result-table number traces to one `tag` below; the parsed per-run metrics and the derived-evidence file accompany this supplement, and full per-run manifests and raw logs are part of the artifact release at publication.

#	Run tag	Category	Status	Note
1	`parquet_conversion`	infra	completed	
2	`block1_stage1_n5_sanity`	sanity	completed	all 3 non-empty windows 100% match
3	`block1_stage1_n50_main`	main	passed	K1 kill-switch PASSED: 90,866 host comparisons across 40 windows, 0 mismatches
4	`block1_stage2_n5_sanity`	sanity	completed	
5	`block1_stage2_n20_main`	main	passed	All 4 Block 1 sub-tests: 100% agreement (point_coreness, core_threshold, core_component,...
6	`b2_tcd_slice1h`	baseline	completed	algorithm time microseconds; wall 1 sec/query due to per-query printf+state overhead
7	`b2_iphc_slice1h`	baseline	completed	iPHC much slower than TCD per query (515 ms vs 0.18 us); confirms TCD/OTCD beats iPHC by...
8	`tracecore_mvp_build_slice1h`	main	completed	naive MVP: in-memory dict-per-epoch; no B+-tree yet
9	`tracecore_mvp_bench_slice1h`	main	completed	point_coreness 0.6x TCD speed; core_threshold 63us; temporal_k_core 536us — beats iPHC 10...
10	`tracecore_redteam_build_28d`	main	completed	28-day full coverage of LANL redteam time range, ready for detection experiments
11	`rolling_fanout_28d`	infra	completed	Per-host sliding window over 11911 hosts; C17693 (dominant attacker) shows clear spikes
12	`detect_redteam_v1`	main	reported	KEY FINDING: pure k-shell scoring gives 0% recall — attackers operate at network PERIPHER...

continued on next page

Table 8 (continued)

#	Run tag	Category	Status	Note
13	detect_redteam_v2	main	completed	: rolling_fanout_72 / (k_shell+1) catches 22.6% of attack epochs at 1pct FP, 63.4pct at 5...
14	detect_nfuq_botiot	baseline	excluded	Dataset too small (293 hosts) and attackers fully separated; all scores AP=1.0
15	detect_nfuq_cicids2018	baseline	completed	KEY MULTI-DATASET RESULT: largest realistic NF-UQ sub-dataset, k_shell gives 24.5pct R@1p...
16	detect_nfuq_toniot	baseline	completed	Method works at AP=0.72 R@1pct=77.4pct on 29K-host IoT testbed; cleaner attacker separati...
17	detect_nfuq_unsw	baseline	excluded	Dataset too small (42 hosts) for meaningful test; fanout and low_shell_fanout AP=1.0
18	supervised_cicids2018	main	completed	Head-to-head with GNN baselines: TraceCore+Flow achieves PR-AUC=0.9983 (vs GraphIDS publi...
19	supervised_toniot	main	completed	+30pp at R@FP=1e-4 from TraceCore features over flow-only baseline; F1 0.9895 matches E-G...
20	supervised_unsw	main	completed	F1 0.9635 beats E-GraphSAGE 0.95 by 1.3pp; graph features useful even in low-attack-rate...
21	supervised_botiot	main	excluded	Dataset 99.6pct attack — too imbalanced for meaningful FP-rate comparison. All methods F1...
22	egraphsage_pure_torch_cicids2018	baseline	baseline	Head-to-head with TraceCore+Flow+LightGBM (F1 0.9816, PR 0.9983, ROC 0.9998, R@1e-3 0.968...
23	egraphsage_pure_torch_toniot	baseline	baseline	Comparison: TraceCore+Flow F1 0.9895 vs E-GraphSAGE 0.7533 -¿ TraceCore +23.6pp F1. Trace...
24	egraphsage_pure_torch_unsw	baseline	baseline	Comparison: TraceCore+Flow F1 0.9635 vs E-GraphSAGE 0.8469 -¿ TraceCore +11.7pp F1. E-Gra...
25	egraphsage_pure_torch_botiot	baseline	excluded	Dataset too imbalanced and graph too small for meaningful comparison; OOM at hidden=16. S...

continued on next page

Table 8 (continued)

#	Run tag	Category	Status	Note
26	graphids_official_cicids2018	baseline	baseline	DIRECT HEAD-TO-HEAD: TraceCore+Flow+LightGBM PR-AUC 0.9983 vs GraphIDS PR-AUC 0.9274 — Tr...
27	supervised_cicids2018_frac0p2	main	completed	APPLES-TO-APPLES vs GraphIDS NeurIPS25 (same fraction=0.2): TraceCore F1 0.9819 vs GraphI...
28	graphids_official_unsw	baseline	baseline	HEAD-TO-HEAD: TraceCore+Flow+LightGBM F1=0.9635 PR=0.9911 vs GraphIDS F1=0.9262 PR=0.7579...
29	supervised_unsw_frac0p2	main	completed	APPLES-TO-APPLES vs GraphIDS NeurIPS25 on UNSW (both fraction=0.2): TraceCore F1 0.9630 v...
30	lanl_role_residual	main	completed	: R@FP=1e-3 went from 0pct (raw) to 78.7pct (rf288 residual); R@FP=1e-2 from 22.6pct to 8...
31	per_attack_class_toniot	main	completed	Per-class FP=1pct: TraceCore +13.2pp on scanning, +35.4pp on mitm, +31.9pp on ransomware...
32	block3_storage_lanl	main	completed	Headline: ShellIndex varint+gzip is 18.73x smaller than naive snapshot uncompressed (or 3...
33	anomale_NF-CSE-CIC-IDS2018-v2	baseline	completed	Anomal-E reimplementation (paper-faithful: E-GraphSAGE encoder + DGI self-supervised + IF...
34	graphids_official_toniot_frac0.2	baseline	completed	GraphIDS (NeurIPS 25) F1 0.6804 / PR-AUC 0.8056 on NF-ToN-IoT-v2 vs TraceCore+LGB F1 0.98...
35	supervised_toniot_frac0.2	main	completed	Apples-to-apples vs GraphIDS-ToN-IoT (F1=0.6804/PR-AUC=0.8055): TraceCore+LGB +30.9pp F1...
36	block1_stage3_oracle	main	completed	R1-M1 fix: extends Block-1 gate to all six primitives. coreness_ascent vs igraph per-epoc...

continued on next page

Table 8 (continued)

#	Run tag	Category	Status	Note
37	`role_residual_r1_sweeps`	ablation	completed	R1-M4: K in 4..32 plateau (R@1e-3 0.756-0.787), collapse at K=64; causal EWMA with full r…
38	`role_residual_r2_deployable`	ablation	completed	R1-M4 corrected: cohort stats causal in all variants; role labels from (i) frozen 500-epo…
39	`protocol_variants_nfuq`	ablation	completed	R1-C2: transductive vs train-only differ by ¡=0.21pp everywhere (paper protocol loses noth…
40	`streaming_systems_bench`	main	completed	R1-M5. W300 run includes concurrent point-query client (270K queries; client snapshots ep…